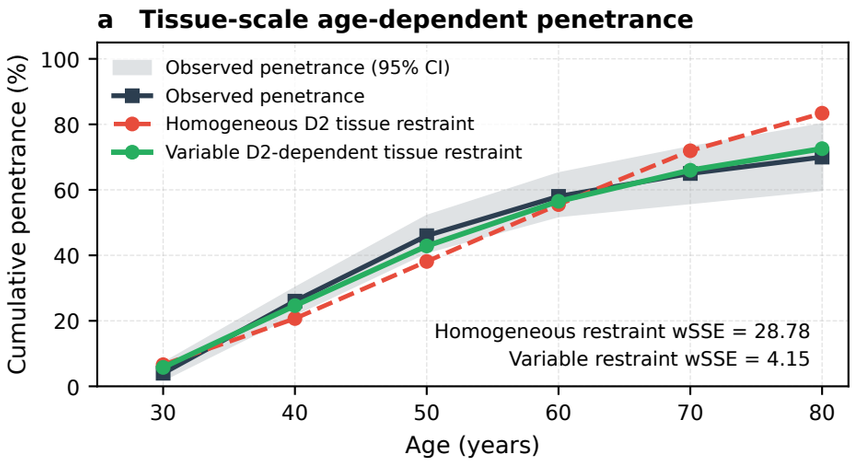


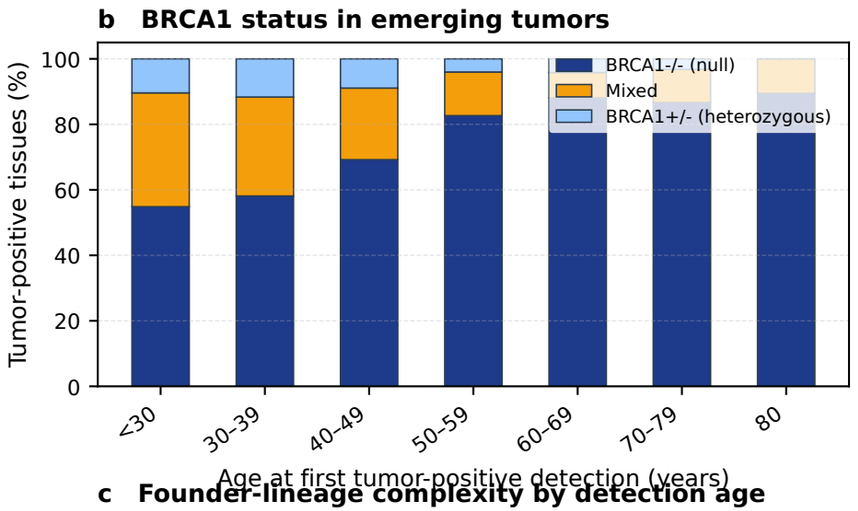


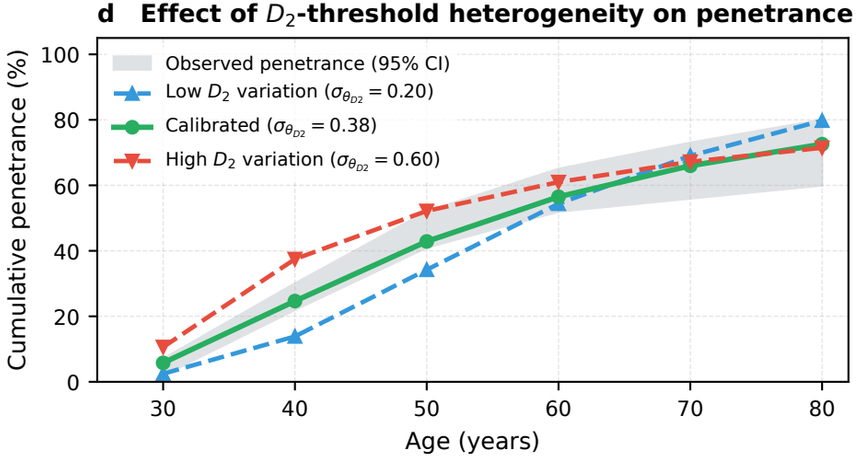


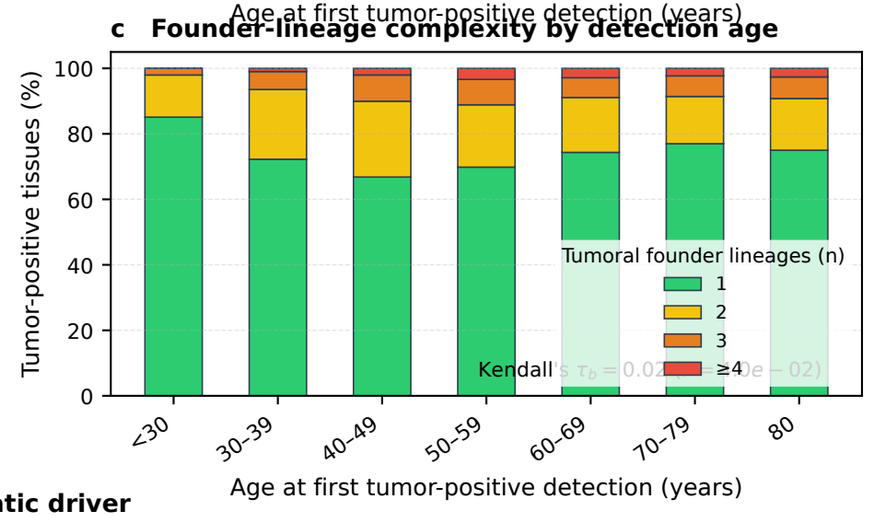


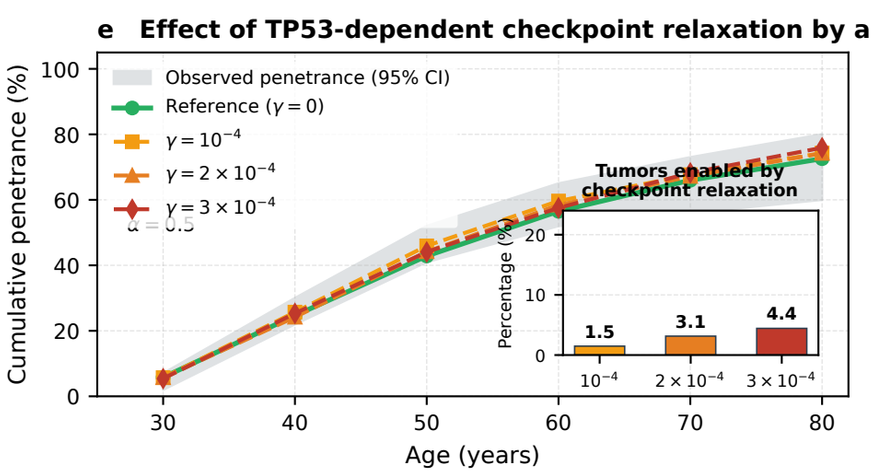


Tissue-Scale Simulation Summary (10^7 cells):

- Calibrated wSSE: 4.15 (vs Kuchenbaecker 2017)
- BRCA1-loss-majority tumors (>50% null): 92.5%
- Subclonal BRCA1+/- retention (≥25%): 21.6%
- Multi-founder tumors (≥2 lineages): 28.1%
- D2 heterogeneity: low sigma delays risk, high sigma accelerates onset
- Checkpoint relaxation: enables alternative event ordering